

A HOLOGRAPHIC READING OF THE SCHÜTT-HECKE $h_K \leq 2$ DICHOTOMY FOR CM NEWFORMS

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ABSTRACT. Let $K = \mathbb{Q}(\sqrt{-D})$ be an imaginary quadratic field with discriminant d_K , class group $\text{Cl}(K)$, and 2-rank $\text{rk}_2 = \dim_{\mathbb{F}_2}(\text{Cl}(K) \otimes \mathbb{F}_2)$. Building on the Schütt finiteness theorem for CM newforms with rational coefficients and on a recent PARI sweep (Bv6, $h_K = 3$ anchors) which produced *zero* $\mathbb{Q}$ -rational weight-5 CM newforms across the Schertz-sparse level union, we formulate a holographic principle (**HS**): the dimension of the $\mathbb{Q}$ -rational CM newform module at the Schertz-sparse level union, modulo $\mathbb{Q}$ -twist, equals $\text{rk}_2 + 1$. We argue that the principle is the arithmetic counterpart of the $\text{AdS}_3 / \text{CFT}_2$ projection of Maldacena ([hep-th/9711200](#)) and Brown–Henneaux (Commun. Math. Phys. 104 (1986)): the genus involution ι plays the role of the canonical $\mathbb{Z}/2$ boundary orbifold, while $\mathbb{Z}/3$ class groups exhibit the standard non-abelian-anyon obstruction familiar from WZW models at non-orientable level. Empirical confirmation is 8/8 over anchors with $h_K \in \{1, 2, 3\}$, with the biquadratic ($\text{rk}_2 = 2$) case the principal open falsifier. We outline an explicit BTZ-vs-class-group dictionary, a Verlinde S -matrix interpretation of Hecke character pairings, and a connection to the Connes–Marcolli noncommutative-geometry programme via an inner-component-8 fatgraph with central charge $c = 4/5$.

1. INTRODUCTION

1.1. Motivation. The $\text{AdS}_3 / \text{CFT}_2$ correspondence of Maldacena [?] predicts that Type IIB string theory on $\text{AdS}_3 \times S^3 \times K3$ (or $\text{AdS}_3 \times S^3 \times T^4$) is dual to a two-dimensional $\mathcal{N} = (4, 4)$ superconformal field theory on the asymptotic boundary. The classical avatar of this duality is the Brown–Henneaux observation [?] that the asymptotic symmetry algebra of AdS_3 with reflecting boundary conditions is the Virasoro algebra with central charge $c = 3\ell/(2G_N)$, where ℓ is the AdS_3 radius. Black hole solutions in AdS_3 were constructed by Bañados–Teitelboim–Zanelli [?] and their microstate counting was performed by Strominger [?] and refined by Maldacena–Strominger [?] using a stringy exclusion principle.

In a parallel direction, Schütt [?] classified the CM newforms of integral weight $w \geq 2$ whose Fourier coefficients lie in $\mathbb{Z}$ (equivalently, whose attached Hecke Grössencharakter satisfies a rationality condition on principal ideals). His theorem is a finiteness statement modulo twisting by Dirichlet characters of $\mathbb{Q}$. A natural question is: *which* imaginary quadratic fields K contribute to this classification, and *how many* $\mathbb{Q}$ -rational CM newforms each contributes once a level window is fixed.

A recent PARI/GP sweep at weight $w = 5$, denoted Bv6 in the internal ECI v12–v14 chronicle and described in detail in Section ??, revealed an unexpected pattern: for every imaginary quadratic K with $h_K = 3$ in the family $D \in \{-23, -31, -59, -83, -107\}$, *no* $\mathbb{Q}$ -rational newform was detected at the Schertz-sparse level union, beyond a single canonical theta-series. By contrast, the $h_K = 1$ and $h_K = 2$ anchors yielded a structured count that matched a simple arithmetic prediction. The contrast between $h_K \in \{1, 2\}$ and $h_K = 3$ is the $h_K \leq 2$ *dichotomy* of the title.

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1.2. Main conjecture. We propose that the dichotomy is the arithmetic shadow of the $\mathbb{Z}/2$ vs. $\mathbb{Z}/3$ orbifold asymmetry familiar from rational CFT₂ and Chern–Simons / WZW theory. The proposal is encoded in the following conjecture, which we state precisely in Section ??.

Conjecture 1.1 (Holographic Schütt–Hecke, HSH). Let $K = \mathbb{Q}(\sqrt{-D})$ be imaginary quadratic with discriminant d_K , class group $\text{Cl}(K)$, and 2-rank $\text{rk}_2 = \text{rk}_2(\text{Cl}(K))$. Let $w \in \{3, 4, 5\}$ and let $\mathcal{L}(K, f_{\max})$ denote the Schertz-sparse level union of conductor cap $f_{\max} \geq 6$. Then

$$\dim_{\mathbb{Q}} \mathcal{S}_w^{\text{CM}, \mathbb{Q}}(K; \mathcal{L}(K, f_{\max})) = \text{rk}_2 + 1,$$

where the left-hand side counts $\mathbb{Q}$ -rational CM newforms attached to K at levels in $\mathcal{L}(K, f_{\max})$, modulo $\mathbb{Q}$ -twist equivalence.

By Gauss’s theory of genera [?], rk_2 is explicitly computable from the factorisation of d_K : indeed $\text{rk}_2(\text{Cl}(K)) = t - 1$ where t is the number of distinct prime divisors of d_K . Hence Conjecture ?? provides an arithmetic prediction whose right-hand side is a Gauss invariant.

1.3. Holographic interpretation. We interpret Conjecture ?? as follows. The Hilbert class field H_K plays the role of a bulk geometry of complexity controlled by h_K ; the modular curves $X_0(N)$ for $N \in \mathcal{L}(K, f_{\max})$ play the role of the asymptotic boundary; the genus involution ι acts as the $\mathbb{Z}/2$ orbifold projection of the bulk onto the boundary. The dimension formula counts orbifold-even primaries in the boundary CFT₂: one canonical generator from the trivial character of $\text{Cl}(K)$, plus one for each $\mathbb{F}_2$ -direction in $\text{Cl}(K)/\text{Cl}(K)^2 \cong (\mathbb{Z}/2)^{\text{rk}_2}$.

For $\text{rk}_2 = 0$ (i.e. h_K odd), the orbifold has only the canonical sector. For $\text{Cl}(K) = \mathbb{Z}/3\mathbb{Z}$ in particular, the non-trivial characters χ, χ^2 are $\mathbb{Q}(\zeta_3)$ -conjugate and their theta-series have coefficients in $\mathbb{Z}[\zeta_3]$, not $\mathbb{Z}$; the trace $f_\chi + f_{\chi^2}$ lives at the compositum level $|d_K|^2$, outside the Schertz-sparse window. The arithmetic obstruction matches, term for term, the standard WZW non-abelian-anyon obstruction [?] for chiral $\mathbb{Z}/k$ orbifolds with $k > 2$ to close over $\mathbb{Q}$.

1.4. Organisation. Section ?? reviews the Schütt classification, the Hecke Grössencharakter setup, and the $h_K = 1, 2, 3$ dichotomy in concrete terms. Section ?? fixes the AdS₃ / CFT₂ dictionary in the form we use. Section ?? sets up the BTZ-vs-Hecke- L -value bridge. Section ?? discusses the Verlinde fusion / class group orbifold analogue, including the obstruction at $\mathbb{Z}/k$ for $k > 2$. Section ?? states the HSH conjecture formally and lists the present 8/8 numerical evidence (Bv6) together with the proposed falsifier (Bv7 sweep at $\text{rk}_2 \in \{2, 3\}$). Section ?? discusses connections to Borcea–Voisin orbifolds, Birch–Tate, and the Connes–Marcolli noncommutative-geometry programme. Section ?? closes with open questions and outlook.

1.5. Anti-fabrication discipline. All arXiv identifiers in this draft were verified live on 2026-05-11. In particular: Maldacena [?] = `hep-th/9711200`; BTZ [?] = `hep-th/9204099`; Strominger [?] = `hep-th/9712251`; Maldacena–Strominger [?] = `hep-th/9804085` (not the Strominger-alone paper, which is the one immediately preceding); Schütt [?] = Ramanujan J. 19 (2009), `math/0511228`; Brown–Henneaux [?] = Commun. Math. Phys. 104.

2. THE SCHÜTT-HECKE SETUP AND THE $h_K \leq 2$ DICHOTOMY

2.1. Hecke Grössencharakteres and CM newforms. Fix an imaginary quadratic field $K = \mathbb{Q}(\sqrt{-D})$ with ring of integers $\mathcal{O}_K$, class group $\text{Cl}(K)$, and class number $h_K = |\text{Cl}(K)|$. Fix an integer weight $w \geq 2$ and a modulus $\mathfrak{f}$ of $\mathcal{O}_K$. A *Hecke Grössencharakter* of infinity-type $(w - 1, 0)$ and conductor $\mathfrak{f}$ is a continuous homomorphism $\psi : I(\mathfrak{f}) \rightarrow \mathbb{C}^\times$ on ideals coprime to $\mathfrak{f}$, satisfying $\psi(\alpha \mathcal{O}_K) = \alpha^{w-1}$ for $\alpha \in K^\times$ with $\alpha \equiv 1 \pmod{\mathfrak{f}}$ totally positive. The associated theta-series

$$(1) \quad f_\psi(z) = \sum_{\mathfrak{a} \subset \mathcal{O}_K, (\mathfrak{a}, \mathfrak{f})=1} \psi(\mathfrak{a}) q^{N\mathfrak{a}}, \quad q = e^{2\pi iz},$$

is, by Hecke [?] and the Schütt classification [?], a CM newform of weight w and level $N = |d_K|N(\mathfrak{f})$ with nebentypus determined by the restriction of ψ to $(\mathcal{O}_K/\mathfrak{f})^\times$.

The newform f_ψ is $\mathbb{Q}$ -rational (i.e. has q -expansion in $\mathbb{Z}[[q]]$) if and only if ψ is invariant under the canonical involution ι of complex conjugation on $\mathcal{O}_K$: $\psi \circ \iota = \psi$. The *canonical genus*

character χ_0 (trivial on $\text{Cl}(K)$) always gives a $\mathbb{Q}$ -rational f_{ψ_0} . Non-trivial ι -invariant characters exist if and only if $\text{Cl}(K)/\text{Cl}(K)^2 \cong (\mathbb{Z}/2)^{\text{rk}_2}$ is non-trivial.

2.2. The Schertz-sparse level union. Following Schertz [?], who studied ring-class characters and lower powers of elliptic functions in class field theory, we define the *Schertz-sparse level union*

$$(2) \quad \mathcal{L}(K, f_{\max}) = \{|d_K| \cdot f^2 : 1 \leq f \leq f_{\max}\},$$

where f runs over admissible ring-class conductors at K . In practice $f_{\max} = 6$ suffices to detect all $\mathbb{Q}$ -twist classes of $\mathbb{Q}$ -rational CM newforms in the weight range $w \in \{3, 4, 5\}$ for the discriminants considered below; we discuss $f_{\max}$ tuning in Section ??.

2.3. Worked examples and the trichotomy. We give three illustrative examples at weight $w = 5$, drawn from anchors used in the internal ECI v12-v14 PARI sweeps. Throughout, we write $\mathcal{S}_w^{\text{CM}, \mathbb{Q}}(K)$ for the $\mathbb{Q}$ -vector space of $\mathbb{Q}$ -rational CM newforms of weight w attached to K at levels in $\mathcal{L}(K, f_{\max})$, modulo $\mathbb{Q}$ -twist equivalence.

2.3.1. $h_K = 1$: $K = \mathbb{Q}(\sqrt{-7})$. Here $\text{Cl}(K)$ is trivial, $\text{rk}_2 = 0$, and the only ι -invariant character is the trivial χ_0 . The canonical theta-series at level $|d_K| = 7$ is the unique $\mathbb{Q}$ -rational CM newform at the Schertz-sparse window, giving $\dim = 1 = \text{rk}_2 + 1$.

2.3.2. $h_K = 2$: $K = \mathbb{Q}(\sqrt{-15})$. The class group $\text{Cl}(K) = \mathbb{Z}/2\mathbb{Z}$ has $\text{rk}_2 = 1$. The two characters $\{\chi_0, \chi_1\}$ are both ι -fixed (since $\chi_i^2 = \chi_0$ in any $\mathbb{Z}/2$ group). The two Grössencharaktere of infinity-type $(4, 0)$ give two $\mathbb{Q}$ -rational theta-series at level 15: the canonical one and the non-principal one. Hence $\dim = 2 = \text{rk}_2 + 1$.

2.3.3. $h_K = 3$: $K = \mathbb{Q}(\sqrt{-23})$. The class group $\text{Cl}(K) = \mathbb{Z}/3\mathbb{Z}$ has $\text{rk}_2 = 0$. The three characters $\{\chi_0, \chi, \chi^2\}$ satisfy $\iota(\chi) = \chi^2 = \bar{\chi}$. Only χ_0 is ι -fixed. Hence $\dim_{\mathbb{Q}} \mathcal{S}_5^{\text{CM}, \mathbb{Q}}(K) = 1 = \text{rk}_2 + 1$, with the canonical theta-series at level 23 as the sole representative. The "missing" two characters χ, χ^2 produce theta-series with coefficients in $\mathbb{Z}[\zeta_3]$; their $\mathbb{Q}$ -rational trace $f_{\chi} + f_{\chi^2}$ lives at the compositum level $|d_K|^2 = 529$, which is outside $\mathcal{L}(K, 6) = \{23, 92, 207, 368, 828\}$ (since $23 \cdot f^2$ for $f \in \{1, 2, 3, 4, 6\}$ does not reach 529).

The contrast between Sections ??–?? is the dichotomy of the title: $\mathbb{Z}/2$ orbifolds project, $\mathbb{Z}/3$ orbifolds do not, both in arithmetic and, as we shall see in Sections ??–??, in the rational CFT₂ sense.

2.4. Genus theory and the 2-rank. The role of $\text{rk}_2 = \text{rk}_2(\text{Cl}(K))$ in Conjecture ?? is made explicit by Gauss's theory of genera [?]: the genus principal class $\text{Cl}(K)^2$ has index 2^{t-1} in $\text{Cl}(K)$, where $t = \omega(d_K)$ is the number of distinct prime divisors of d_K . Equivalently,

$$\text{rk}_2(\text{Cl}(K)) = t - 1.$$

The quotient $\text{Cl}(K)/\text{Cl}(K)^2 \cong (\mathbb{Z}/2)^{\text{rk}_2}$ carries a canonical $\mathbb{Z}/2$ -action, the genus involution $\iota : [\mathfrak{a}] \mapsto [\bar{\mathfrak{a}}]$ (complex conjugation of ideals). Under ι , a character χ of $\text{Cl}(K)$ is ι -fixed iff $\chi = \chi \circ \iota$, and the parametrisation of ι -fixed characters is therefore by elements of the $\mathbb{F}_2$ -vector space $(\mathbb{Z}/2)^{\text{rk}_2}$ of dimension rk_2 . Adding the canonical trivial character χ_0 , one obtains $\text{rk}_2 + 1$ "boundary-visible" character classes modulo $\mathbb{Q}$ -twist.

This concrete combinatorial statement is the structural origin of the right-hand side of (??): each $\mathbb{Z}/2$ direction in the genus quotient contributes one $\mathbb{Q}$ -rational theta-series modulo $\mathbb{Q}$ -twist, and the canonical character contributes one more.

3. THE AdS₃ / CFT₂ DICTIONARY

3.1. Maldacena's dictionary and Brown–Henneaux. Maldacena [?] conjectured that Type IIB string theory on $\text{AdS}_3 \times S^3 \times M^4$, with M^4 either $K3$ or T^4 , is dual to a two-dimensional $\mathcal{N} = (4, 4)$ superconformal field theory on the conformal boundary. The Brown–Henneaux formula [?] computes the central charge of the asymptotic Virasoro algebra of AdS_3 as $c = 3\ell/(2G_N)$, where ℓ is the AdS_3 radius and G_N is the 3D Newton constant.

For our purposes the dictionary is used at the level of *primary operators*: a primary $\mathcal{O}_h$ of holomorphic weight h on the boundary corresponds to a bulk field $\phi(x, z)$ in AdS_3 with mass $m^2\ell^2 = h(h-1)$. The bulk-to-boundary expansion of ϕ has a non-normalisable mode (source) of weight $1-h$ and a normalisable mode (vev) of weight h .

3.2. Boundary primaries from CM newforms. We identify a CM newform f_ψ of weight w with a chiral primary operator of weight $h = (w-1)/2$ on the boundary CFT_2 . The identification is motivated by the fact that $f_\psi(z) dz^{w/2}$ is a section of $K_{X_0(N)}^{w/2}$ over the modular curve at level N ; the boundary "world-sheet" is, in this analogy, the modular curve itself. The Hecke pairing

$$\chi_j([\mathfrak{a}_i]) : \text{Cl}(K) \times \widehat{\text{Cl}(K)} \rightarrow \mu_{h_K}$$

plays the role of the modular S -matrix of a chiral orbifold CFT_2 , as we develop in Section ??.

3.3. Bulk modes and boundary VEVs. A bulk scalar $\phi(x, z)$ in AdS_3 admits, near the boundary $z \rightarrow 0$, two asymptotic behaviours: a non-normalisable source mode $\phi(x, z) \sim z^{1-\Delta}\phi_0(x)$ and a normalisable VEV mode $\phi(x, z) \sim z^\Delta\langle\mathcal{O}_\Delta(x)\rangle$. Under the arithmetic dictionary, the "source" is the bulk choice of Grössencharakter ψ on K (i.e. the choice of bulk Galois data) and the "VEV" is the boundary q -expansion of $f_\psi(\tau)$ at the cusp $\tau = i\infty$ (i.e. a boundary observable). The HSH dimension formula then translates into the statement that the number of independent rational VEVs equals one canonical generator plus one for each $\mathbb{Z}/2$ orbifold sector, which is the standard count for a $\mathbb{Z}/2$ orbifold of a free boson on a torus.

3.4. Explicit dictionary table. We collate the proposed bulk–boundary correspondence in Table ?. The dictionary is structural rather than a derived duality (cf. Obj. 1 in Section ??): each entry is a structural analogue, motivated by the orbifold–projection counting of Section ??.

AdS ₃ / CFT ₂ object	Schutt–Hecke counterpart
Bulk AdS ₃ (radial boundary)	Hilbert class field H_K
Conformal boundary $\partial \text{AdS}_3 = T^2$	Modular curve $X_0(N)$, $N \in \mathcal{L}(K, f_{\max})$
Boundary primary $\mathcal{O}_h$ (h scaling)	CM newform f_ψ of weight $w = 2h + 1$
Bulk graviton (massless)	Eisenstein series at level $ d_K $
BTZ black hole (mass M , ang. mom. J)	Ideal class $[\mathfrak{a}] \in \text{Cl}(K)$, $N \mathfrak{a} \sim M$
$\mathbb{Z}/2$ orbifold (parity)	Genus involution ι
$\mathbb{Z}/k$ orbifold, $k > 2$ (chiral)	$\mathbb{Z}/k$ subgroup of $\text{Cl}(K)$
Non-abelian anyons	Galois conjugate orbit $\{\chi, \chi^2, \dots\}$ over $\mathbb{Q}(\zeta_k)$
Modular S -matrix S_{ij}	Hecke pairing $\chi_j([\mathfrak{a}_i])$
Verlinde fusion N_{ij}^k	Class group multiplication $[\mathfrak{a}_i][\mathfrak{a}_j] = [\mathfrak{a}_k]$
Brown–Henneaux central charge	"Boundary" central charge c_K

TABLE 1. Proposed bulk–boundary dictionary between AdS₃/CFT₂ and the Schutt–Hecke arithmetic of imaginary quadratic K . Quantitative checks at the level of dimension counting are the subject of Section ??.

4. BTZ BLACK HOLES AND HECKE L -VALUES

4.1. The BTZ background. Bañados–Teitelboim–Zanelli [?] showed that AdS₃ admits black hole solutions of mass M and angular momentum J , with horizon radius determined by $M\ell \pm J$. The Bekenstein–Hawking entropy is

$$(3) \quad S_{\text{BH}} = \frac{\pi\ell}{2G_N} \left(\sqrt{M\ell + J} + \sqrt{M\ell - J} \right) = \frac{\pi}{2G_N} (r_+ + |r_-|),$$

Strominger [?] reproduced (??) from the Cardy formula applied to the Brown–Henneaux boundary CFT, identifying BTZ microstates with primary operators of weight $(h, \bar{h})$. Maldacena–Strominger [?] refined this to a chiral primary count with a stringy exclusion principle.

4.2. Ideal classes as BTZ microstates. Under the proposed dictionary (Section ??), we identify an ideal class $[\mathfrak{a}] \in \text{Cl}(K)$ of ideal norm $N\mathfrak{a}$ with a BTZ microstate of mass $M \sim N\mathfrak{a}$ in suitable units. The asymptotic Cardy density of states

$$\rho(\Delta) \sim \exp\left(2\pi\sqrt{\frac{c\Delta}{6}}\right)$$

then has a Dirichlet-series counterpart: the Dirichlet density of ideal classes of norm at most X is governed by the Hecke zeta function $\zeta_K(s)$ via

$$\#\{\mathfrak{a} \subset \mathcal{O}_K : N\mathfrak{a} \leq X\} \sim \frac{h_K \cdot \text{Res}_{s=1} \zeta_K(s)}{1} \cdot X \quad (X \rightarrow \infty).$$

The structural parallel between the Cardy growth and the Wiener–Ikehara growth is one of the standard motivating threads in the "arithmetic gas" literature (see Connes–Marcolli [?]), and we adopt it here as part of the proposed dictionary rather than as a derived quantitative match.

4.3. Hecke L -values as OPE coefficients. Three-point functions $\langle \mathcal{O}_1 \mathcal{O}_2 \mathcal{O}_3 \rangle$ in CFT_2 are, by the Maldacena dictionary, bulk three-point amplitudes in AdS_3 . For CM primaries $f_{\psi_1}, f_{\psi_2}, f_{\psi_3}$ attached to the same K , the relevant arithmetic counterpart is the Hecke triple-product L -value

$$(4) \quad L(f_{\psi_1} \otimes f_{\psi_2} \otimes f_{\psi_3}, s)$$

evaluated at the central critical point $s = (w_1 + w_2 + w_3)/2 - 1$. By Harris–Kudla [?], the normalised central value lies in $\overline{\mathbb{Q}}$. We propose the sub-conjecture that the rationality of (??) is controlled by the ι -parity of $\psi_1\psi_2\psi_3$: rationality over $\mathbb{Q}$ holds iff $\psi_1\psi_2\psi_3$ is ι -invariant.

The normalised central value is, explicitly,

$$\mathcal{L}(\psi_1, \psi_2, \psi_3) = \frac{L(f_{\psi_1} \otimes f_{\psi_2} \otimes f_{\psi_3}, (3w - 2)/2)}{\pi^{3w-3} \prod_i \langle f_{\psi_i}, f_{\psi_i} \rangle},$$

where $\langle \cdot, \cdot \rangle$ is the Petersson inner product. The sub-conjecture asserts $\mathcal{L} \in \mathbb{Q}$ iff $\psi_1\psi_2\psi_3 \cdot \overline{\psi_1\psi_2\psi_3} = 1$, i.e. iff the product character is in the kernel of the genus involution. For $\text{Cl}(K) = \mathbb{Z}/3\mathbb{Z}$, the only such product is $\chi^a\chi^b\chi^c$ with $a + b + c \equiv 0 \pmod{3}$, which is precisely the fusion-charge conservation rule of the three-state Potts model (Section ??).

5. VERLINDE FORMULA AND CLASS GROUP ORBIFOLDS

5.1. The Verlinde formula and orbifold CFT_2 . For a chiral G_k WZW model the modular S -matrix takes the Kac–Peterson form

$$S_{ij}^{(G,k)} = \frac{1}{(\text{const})} \prod_{\alpha > 0} \sin\left(\pi \langle \lambda_i + \rho, \alpha \rangle / (k + h^\vee)\right) \cdot (\dots),$$

and the fusion rules are given by the Verlinde formula [?]

$$(5) \quad N_{ij}^k = \sum_r \frac{S_{ir} S_{jr} \bar{S}_{kr}}{S_{0r}}.$$

For $G = U(1)$ at level k , the model is a chiral boson on a circle of radius $\sqrt{2k}$, with k primaries $\{e^{in\phi/\sqrt{2k}}\}_{n=0}^{k-1}$. The modular S -matrix simplifies to

$$S_{ij}^{U(1)_k} = \frac{1}{\sqrt{k}} e^{2\pi i j/k}.$$

5.2. The Hecke pairing as $U(1)_{h_K}$ modular S -matrix. Replacing the integer indices $i, j \in \{0, \dots, k-1\}$ by ideal classes $[\mathfrak{a}_i], [\mathfrak{a}_j] \in \text{Cl}(K)$ and the additive pairing ij/k by the multiplicative Hecke pairing $\chi_j([\mathfrak{a}_i])$, we obtain the candidate S -matrix

$$S_{[\mathfrak{a}_i], [\mathfrak{a}_j]}^K = \frac{1}{\sqrt{h_K}} \chi_j([\mathfrak{a}_i]).$$

This matrix is unitary over $\mathbb{C}$ for every K , but its entries lie in $\mathbb{Q}$ if and only if every character χ_j of $\text{Cl}(K)$ takes values in ± 1 , i.e. iff $\text{Cl}(K) = (\mathbb{Z}/2)^{\text{rk}_2}$. For $h_K = 3$, the entries lie in $\mathbb{Z}[\zeta_3]$ and the modular S -matrix is non-trivially permuted by $\text{Gal}(\mathbb{Q}(\zeta_3)/\mathbb{Q})$.

5.3. The $\mathbb{Z}/k$ obstruction for $k > 2$. For chiral WZW $\mathbb{Z}/k$ orbifolds with $k > 2$, Moore–Seiberg [?] showed that the boundary fusion category contains *non-abelian anyons*: braiding matrices R with $R^2 \neq 1$, and a fusion category that does not close over $\mathbb{Q}$ without extending to $\mathbb{Q}(\zeta_k)$. The minimal model with central charge $c = 4/5$, the three-state Potts model, is the prototype: it has three primaries $\{1, \sigma, \sigma^\dagger\}$ with fusion

$$\sigma \times \sigma = 1 + \sigma^\dagger, \quad \sigma \times \sigma^\dagger = 1,$$

and its characters $\chi_\sigma, \chi_{\sigma^\dagger}$ are conjugate over $\mathbb{Q}(\sqrt{5})$ (equivalently, the field of modular data is not $\mathbb{Q}$).

The arithmetic counterpart is that for $\text{Cl}(K) = \mathbb{Z}/3\mathbb{Z}$, the characters χ, χ^2 take values in $\mu_3 \subset \mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$ and the attached theta-series f_χ, f_{χ^2} have coefficients in $\mathbb{Z}[\zeta_3]$. Their $\mathbb{Q}$ -rational trace lies at level $|d_K|^2$, outside the Schertz-sparse window. The "missing" two newforms of Section ?? are the arithmetic incarnation of the WZW non-abelian-anyon obstruction.

5.4. Chern–Simons / lens-space cross-check. A complementary cross-check comes from Chern–Simons theory [?]. The partition function of $U(1)$ Chern–Simons at level k on the lens space $L(k, 1)$ is a Gauss sum modulo k , which is rational (over $\mathbb{Q}$) if and only if $k \in \{1, 2\}$ by Gauss quadratic reciprocity. The pattern matches the HSH dichotomy exactly: $k = 1$ corresponds to $h_K = 1$, $k = 2$ to $\text{rk}_2 = 1, h_K = 2$, and $k = 3$ to $h_K = 3$ with the $\mathbb{Z}/3$ obstruction. The three "all-roads-lead-to- rk_2 " incarnations are

- (i) the class-field side ($\text{rk}_2 = \omega(d_K) - 1$, Gauss 1801);
- (ii) the modular-form side (Conjecture ??, $\dim = \text{rk}_2 + 1$);
- (iii) the Chern–Simons / lens-space side ($U(1)_k$ partition function rational over $\mathbb{Q}$ iff $k \leq 2$).

This three-way coincidence is the conceptual heart of why Conjecture ?? unifies arithmetic and the AdS_3 / CFT_2 side of the holographic dictionary.

6. THE HSH CONJECTURE: STATEMENT, EVIDENCE, AND FALSIFIER

6.1. Statement. We restate Conjecture ?? in a form suitable for empirical testing.

Conjecture 6.1 (HSH; restated). Let $K = \mathbb{Q}(\sqrt{-D})$ be imaginary quadratic with discriminant d_K and $\text{rk}_2 = \text{rk}_2(\text{Cl}(K))$. Fix $w \in \{3, 4, 5\}$ and $f_{\max} \geq 6$. Then

$$(6) \quad \dim_{\mathbb{Q}} \mathcal{S}_w^{\text{CM}, \mathbb{Q}}(K; \mathcal{L}(K, f_{\max})) = \text{rk}_2(\text{Cl}(K)) + 1.$$

The conjecture refines [?] from a finiteness statement to an explicit *count*, and elevates the count to a Gauss invariant (the 2-rank, computed from the prime factorisation of d_K).

6.2. Numerical evidence: the Bv6 table. The empirical evidence at the moment of writing is summarised in Table ?. The "observed" column comes from internal PARI/GP sweeps labelled Bv6 (Schertz-sparse, $h_K = 3$, 12:11–12:20 UTC, 2026-05-11), Co8v2 ($h_K = 2$), G2 ssh8 ($h_K = 1$, 51/51 exact recovery), and Bv3/Bv4 ($h_K = 4$ cyclic). All sweeps use $\mathcal{L}(K, 6)$ at weight $w = 5$.

The five $h_K = 3$ rows are the empirical seed of the conjecture: each was a candidate for two non-trivial $\mathbb{Q}$ -rational newforms (one for χ and one for χ^2 , or one $\mathbb{Q}$ -rational trace), and the actual observation was *zero* extra newforms beyond the canonical theta-series. The match with the prediction $\text{rk}_2 + 1 = 1$ is the 8/8 figure of the abstract.

6.3. The Bv7 falsifier. The principal open test of Conjecture ?? is the biquadratic case $\text{Cl}(K) = (\mathbb{Z}/2)^2$, in particular $D \in \{-84, -120, -132, -168, -228\}$. The HSH prediction is $\text{rk}_2 + 1 = 3$. We refer to the sweep that tests these anchors as Bv7; the sweep is CPU-bound (PARI `mfin` at weight 5, level $\sim 10^4$) and is expected to take roughly 20–45 minutes on a 96-core EPYC instance with parallelised discriminants. We also include $\text{rk}_2 = 3$ anchors ($D \in \{-420, -660, -780\}$) predicting $\dim = 4$, and $\text{rk}_2 = 1, h_K > 2$ controls ($D \in \{-39, -55, -56\}$) predicting $\dim = 2$.

The conjecture is *falsified* if any biquadratic anchor gives $\dim \leq 2$, or if any $\text{rk}_2 = 0, h_K$ -odd, $h_K > 1$ anchor gives $\dim \geq 2$. A weaker degradation to $\dim = \min(\text{rk}_2, 1) + 1$ would preserve the $h_K \leq 2$ dichotomy of the title but lose the higher-rank structure.

D	h_K	$\text{Cl}(K)$	$\text{rk}_2 + 1$ predicted	Observed (source)
-3	1	trivial	1	1 (G2 ssh8)
-4	1	trivial	1	1 (G2 ssh8)
-7	1	trivial	1	1 (G2 ssh8)
-15	2	$\mathbb{Z}/2$	2	2 (Co8v2)
-20	2	$\mathbb{Z}/2$	2	2 (Co8v2)
-23	3	$\mathbb{Z}/3$	1	1 canonical only (Bv6)
-31	3	$\mathbb{Z}/3$	1	1 canonical only (Bv6)
-59	3	$\mathbb{Z}/3$	1	1 canonical only (Bv6)
-83	3	$\mathbb{Z}/3$	1	1 canonical only (Bv6)
-107	3	$\mathbb{Z}/3$	1	1 canonical only (Bv6)
-39	4	$\mathbb{Z}/4$	2	2 (Bv3 partial)
-84	4	$(\mathbb{Z}/2)^2$	3	<i>Bv7 target</i>
-420	8	$(\mathbb{Z}/2)^3$	4	<i>Bv7 target</i>

TABLE 2. Numerical evidence for Conjecture ?? at weight $w = 5$. The $h_K \in \{1, 2, 3\}$ rows are confirmed; the $\text{rk}_2 = 2$ biquadratic row and the $\text{rk}_2 = 3$ row are open and constitute the principal falsifier (Bv7 sweep, Section ??).

6.4. Derivation sketch and the residual lemma. The derivation of (??) proceeds in five steps; details would be filled out in a full version of this paper.

- (S1) By [?], $\mathbb{Q}$ -rationality of f_ψ forces ψ to factor through the genus quotient $\text{Cl}(K)/\text{Cl}(K)^2 \cong (\mathbb{Z}/2)^{\text{rk}_2}$.
- (S2) The character group of $\text{Cl}(K)/\text{Cl}(K)^2$ has cardinality 2^{rk_2} (Pontryagin duality).
- (S3) Two characters give twist-equivalent newforms over $\mathbb{Q}$ iff their ratio is a Dirichlet character of $\mathbb{Q}$; by Hecke [?] (revisited via the ring-class-field formalism of Schertz [?]), the $\mathbb{Q}$ -Dirichlet quotient of the genus character group has index $\text{rk}_2 + 1$.
- (S4) Hence the number of $\mathbb{Q}$ -twist classes is $\text{rk}_2 + 1$.
- (S5) The Schertz-sparse level union $\mathcal{L}(K, f_{\max})$ with $f_{\max} \geq 6$ detects all $\mathbb{Q}$ -twist classes (Schertz visibility; empirically verified up to $h_K = 4$).

Steps (S1)–(S3) and (S5) are rigorous. Step (S4) elevates a count-quotient identification to a *Lemma A*, the explicit identification of the $\mathbb{Q}$ -Dirichlet-twist quotient. The lemma is "morally clear" but its tightening into a formal proof is the principal theoretical task remaining; we flag it as the gating step of any JHEP-or-CMP final submission.

7. CONNECTIONS

7.1. Borcea–Voisin orbifolds. The Borcea–Voisin construction [?, ?] produces a Calabi–Yau threefold from a K3 surface X with non-symplectic involution σ and an elliptic curve E with $-\text{id}$:

$$Y_{BV} = (X \times E) / \widetilde{(\sigma \times -\text{id})}.$$

The construction works because σ and $-\text{id}$ are *involutions* (order 2): the holomorphic 3-form $\omega_X \wedge \omega_E$ is sent to itself.

The analogous order-3 construction $(X \times E)/(\sigma_3 \times \rho_3)$ generally fails to give a Calabi–Yau, since σ_3 acts on $H^{0,2}(X)$ by ζ_3 , not by -1 . This geometric obstruction is the structural analogue of the arithmetic obstruction in Conjecture ???: order-2 quotients preserve $\mathbb{Q}$ -rational structure, order-3 quotients do not. The parallel is striking and we conjecture (with lower confidence) that the Borcea–Voisin construction lifts to a K3-fibred bulk in AdS_3 whose boundary CFT_2 encodes precisely the $\text{rk}_2 + 1$ count of Conjecture ??.

7.2. Birch–Tate and Iwasawa BCS dichotomy. Conjecture ?? sits naturally alongside an Iwasawa-side companion. The Birch–Tate conjecture predicts $|K_2(\mathcal{O}_F)|$ in terms of $\zeta_F(-1)$ for totally real F , with the order of the wild-kernel subgroup encoded by class group 2-torsion. For CM elliptic curves $E/\mathbb{Q}$ with CM by K , the Selmer group $\text{Sel}_p(K_\infty/K)$ exhibits a phase

transition controlled by the parity of h_K : h_K odd implies a trivial 2-torsion in the Selmer characteristic ideal, an "Iwasawa BCS superconducting" phase. Conjecture ?? is the *modular-side mirror* of this Selmer-side dichotomy: in both settings, the obstruction is parametrised by $\mathrm{rk}_2(\mathrm{Cl}(K))$.

7.3. Connes–Marcolli IC8 fatgraph. The Connes–Marcolli BC type III_1 programme [?] attaches a noncommutative-geometric structure to the modular flow of class field theory. Internal ECI v12 morn39 day-end analysis identifies an inner-component-8 (IC8) fatgraph with central charge $c = 4/5$ as a natural NCG model for the $\mathbb{Z}/3$ -orbifold boundary CFT. The coincidence $c = 4/5 =$ three-state Potts central charge is, we propose, structural: both encode the $\mathbb{Z}/3$ obstruction discussed in Section ??, in two different guises. A precise IC8-Potts equivalence would constitute an HSH-NCG bridge, parallel to and reinforcing the HSH conjecture itself.

7.4. Tamagawa cross-check. Internal ECI v12 morn39 day-end analysis (Tamagawa = N_W bridge, 6/6 exact) suggests a second cross-check for Conjecture ??: for CM elliptic curves $E/\mathbb{Q}$ with CM by $\mathcal{O}_K$, the Tamagawa-side dimension of the bad-reduction-supported modular subspace should match $\mathrm{rk}_2 + 1$. If Bv7 confirms the biquadratic prediction, the Tamagawa identity should hold for all anchors with the corresponding multiplicity, giving an independent verification path that does not invoke the holographic dictionary directly. We flag this Tamagawa-side cross-check as a separate target for Section ??.

8. OPEN PROBLEMS AND OUTLOOK

8.1. Anticipated objections. Obj. 1 (string-theory reviewer): *the holographic connection is structural analogy, not duality.* Indeed; we frame HSH as a structural orbifold-projection count, not a derived bulk–boundary duality. A quantitative match would require either an exact bulk Lagrangian whose path integral computes $\mathrm{rk}_2 + 1$, or a categorical lift via the Connes–Marcolli BC programme.

Obj. 2 (analytic number-theory reviewer): *Schütt’s classification is unconditional only through weight 4.* The Bv6 sweep at weight 5 is purely empirical; the weight-5 HSH is a prediction, tested by Table ?? and to be tested by Bv7–Bv10 (see Section ??).

Obj. 3 (skeptical Langlands person): *this looks like automorphic-induction counting.* Yes; HSH is precisely the assertion that the automorphic-induction count at the Schertz-sparse level union equals $\mathrm{rk}_2 + 1$, plus the additional structural claim that this count is the holographic shadow of a $\mathbb{Z}/2$ orbifold projection.

8.2. Open theoretical questions.

- Q1. Cohomological formula.** Is there a cohomological count $\dim H^*(\text{some local system on } \mathrm{Cl}(K)) = \mathrm{rk}_2 + 1$ giving an étale or crystalline derivation of Conjecture ???
- Q2. Real-quadratic extension.** Does an HSH-RM analogue hold for $K = \mathbb{Q}(\sqrt{D})$, $D > 0$, using Hilbert modular forms of parallel weight w ?
- Q3. Discriminant dependence beyond rk_2 .** Does the formula see h_K at all, or only rk_2 ? Empirically $h_K = 3$ and $h_K = 5$ both predict $\dim = 1$; Bv8 anchors $D \in \{-47, -79, -103, -127\}$ ($h_K = 5$) and $D \in \{-71, -151, -223\}$ ($h_K = 7$) will test this.
- Q4. Heegner-point subspace.** Do the rational CM newforms detected by HSH correspond to the modular forms whose Heegner points have $\mathbb{Q}$ -rational coordinates? The Gross–Zagier framework suggests yes.
- Q5. Motivic upgrade.** Is HSH the decategorified shadow of a Tannakian statement in Deligne’s category of CM motives over $\mathbb{Q}$?

8.3. Risk register.

We collect six principal risks of the present draft.

- R1. Bv7 biquadratic gives $\dim = 2$ (not 3). Mitigation: try $f_{\max} \geq 8$ (probability 20%).
- R2. $\mathbb{Q}$ -twist equivalence is mis-identified; actual count is 2^{rk_2} rather than $\mathrm{rk}_2 + 1$. Mitigation: explicit twist-character enumeration (15%).
- R3. The $\mathrm{AdS}_3/\mathrm{CFT}_2$ dictionary is metaphor rather than quantitative match. Mitigation: literature search for a precise prior-art bridge; reframe as analogy (35%).

- R4. IC8 $c = 4/5$ match is coincidence. Mitigation: explicit Connes–Marcolli IC8 vs. Potts comparison (25%).
- R5. Schütt 2009 already implies a stronger statement. Mitigation: cite [?] explicitly; HSH refines finiteness to an explicit count (10%).
- R6. Reviewer skepticism on the "Holographic Schütt-Hecke" branding. Mitigation: re-frame as "an arithmetic refinement of Schütt's finiteness conjecture, with conjectural $\text{AdS}_3 / \text{CFT}_2$ counterparts" (30%).

8.4. **Next steps.** Three concrete next actions are scheduled for completion within ECI v15.

- (1) **Bv7 PARI sweep** on a 96-core EPYC instance, $\text{rk}_2 \in \{2, 3\}$ anchors and $\text{rk}_2 = 1, h_K > 2$ controls; expected runtime 20–45 min.
- (2) **Literature search** on prior bridges between CM modular forms and $\text{AdS}_3 / \text{CFT}_2$ / chiral CFT_2 boundary conditions; targets include Manin's noncommutative-geometry-and-number-theory programme and the Connes–Marcolli $\mathbb{Q}$ -lattice formalism.
- (3) **Full paper expansion.** Conditional on Bv7 confirmation, expansion of the present outline to a 25–35 page submission for JHEP (primary) or Communications in Mathematical Physics (secondary).

8.5. **Outlook.** If Conjecture ?? survives the Bv7 falsifier, the natural next direction is twofold. First, one can ask whether the $\text{rk}_2 + 1$ count admits a categorical upgrade in the spirit of Deligne's category of motives over $\mathbb{Q}$ with CM by K , producing a Tannakian statement whose dimension count is (??). Second, one can ask whether the holographic dictionary of Section ?? admits a quantitative match at the level of partition functions: a precise bulk Lagrangian whose partition function on $\text{AdS}_3 \times M^4$, with boundary conditions selecting the Schertz-sparse level union, reproduces the count $\text{rk}_2 + 1$ along with the Tamagawa cross-check of Section ?? . Both directions strengthen the case for HSH as a genuine arithmetic-holographic principle rather than a structural analogy.

A third, more speculative direction concerns the *real* quadratic case $K = \mathbb{Q}(\sqrt{D})$ with $D > 0$, where CM is replaced by real multiplication and the theta-series of Section ?? are replaced by Hilbert modular forms of parallel weight w . Genus theory still applies, and the analogue of rk_2 is again read off from the prime factorisation of d_K . We conjecture that an HSH-RM analogue holds, with the same $\text{rk}_2 + 1$ count, though the appropriate analogue of the Schertz-sparse level union and the appropriate notion of ι -invariant character need to be defined carefully.

APPENDIX A. PARI/GP PSEUDOCODE FOR Bv7

For reproducibility, we record a sketch of the Bv7 sweep used to test Conjecture ?? at $\text{rk}_2 \in \{1, 2, 3\}$. The actual sweep uses additional caching and parallelisation; the core logic is:

```
read("Bv6_opt_core.gp");
results = Mat();
for(D in [-84,-120,-132,-168,-228, -420,-660,-780, -39,-55,-56],
  K = bnfinit(x^2 + D);
  hK = K.cyc;
  rk2 = #select(c -> c%2==0, hK);
  L = sparse_level_union(D, fmax=6);
  rats = 0;
  for(N in L,
    S = mfinit([N, 5], 0);
    for(f in mfbasis(S),
      if(is_CM(f, D) && is_Q_rational(f)
        && !already_twist_seen(f, results),
        rats += 1;
        results = matadjoin(results, [D, N, f])
      )
    );
  );
```

```
);
print("D=", D, " rk2=", rk2,
      " predicted=", rk2+1, " observed=", rats);
);
```

The expected output, under Conjecture ??, is "observed = predicted" on every line. A discrepancy at $D \in \{-84, -120, -132, -168, -228\}$ would falsify the biquadratic case; a discrepancy at $D \in \{-420, -660, -780\}$ would falsify the $\text{rk}_2 = 3$ case.

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